Day 081 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Use Double Number Lines

Standards: 6.RP.A.3a

Learning target: I can represent a ratio on a double number line.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Place zero on both parallel number lines. Align a known pair above and below one tick. Add equal jumps on each line so aligned values remain in the same ratio.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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01 / Learn

Place zero on both parallel number lines. Align a known pair above and below one tick. Add equal jumps on each line so aligned values remain in the same ratio.

Worked example 1

Draw aligned number lines for 3 liters of paint covering 5 square meters. Mark 0, one batch, and 6 batches. How much area do 18 liters cover?

1. Place zero on both parallel number lines.

2. Align a known pair above and below one tick.

3. 18 ÷ 3 = 6 batches

4. 6 × 5 = 30.

5. State the result in the form requested: 30 square meters; aligned pairs (0,0), (3,5), (18,30).

Answer: 30 square meters; aligned pairs (0,0), (3,5), (18,30)

Worked example 2

Draw a double number line for 5 liters of paint covering 4 m². Part A: align 0, 1, 2, and 4 batches. Part B: enter the area covered by 20 liters and explain why the scales can be different.

1. Place zero on both parallel number lines.

2. Each aligned jump adds 5 liters and 4 m².

3. Both quantities scale by 4

4. the number lines measure different units.

Answer: Aligned pairs (0,0), (5,4), (10,8), (20,16); 16 m²

Day 081 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw aligned number lines for 3 liters of paint covering 7 square meters. Mark 0, one batch, and 5 batches. How much area do 15 liters cover?

02 / Constructed response

Draw aligned number lines for 8 liters of paint covering 2 square meters. Mark 0, one batch, and 3 batches. How much area do 24 liters cover?

03 / Constructed response

Draw aligned number lines for 2 liters of paint covering 7 square meters. Mark 0, one batch, and 6 batches. How much area do 12 liters cover?

04 / Constructed response

Draw aligned number lines for 4 liters of paint covering 3 square meters. Mark 0, one batch, and 5 batches. How much area do 20 liters cover?

05 / Constructed response

Draw aligned number lines for 6 liters of paint covering 5 square meters. Mark 0, one batch, and 5 batches. How much area do 30 liters cover?

06 / Constructed response

Draw aligned number lines for 4 liters of paint covering 3 square meters. Mark 0, one batch, and 2 batches. How much area do 8 liters cover?

07 / Constructed response

Draw aligned number lines for 6 liters of paint covering 2 square meters. Mark 0, one batch, and 4 batches. How much area do 24 liters cover?

08 / Double number line and explanation

Draw a double number line for 6 liters of paint covering 4 m². Part A: align 0, 1, 2, and 7 batches. Part B: enter the area covered by 42 liters and explain why the scales can be different.

09 / Double number line and explanation

Draw a double number line for 7 liters of paint covering 6 m². Part A: align 0, 1, 2, and 6 batches. Part B: enter the area covered by 42 liters and explain why the scales can be different.

10 / Constructed response

Draw aligned number lines for 6 liters of paint covering 7 square meters. Mark 0, one batch, and 4 batches. How much area do 24 liters cover?

Day 081 | Unit 06 | Answer key

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1. 35 square meters; aligned pairs (0,0), (3,7), (15,35)

15 ÷ 3 = 5 batches; 5 × 7 = 35.

6.RP.A.3a | Constructed response

2. 6 square meters; aligned pairs (0,0), (8,2), (24,6)

24 ÷ 8 = 3 batches; 3 × 2 = 6.

6.RP.A.3a | Constructed response

3. 42 square meters; aligned pairs (0,0), (2,7), (12,42)

12 ÷ 2 = 6 batches; 6 × 7 = 42.

6.RP.A.3a | Constructed response

4. 15 square meters; aligned pairs (0,0), (4,3), (20,15)

20 ÷ 4 = 5 batches; 5 × 3 = 15.

6.RP.A.3a | Constructed response

5. 25 square meters; aligned pairs (0,0), (6,5), (30,25)

30 ÷ 6 = 5 batches; 5 × 5 = 25.

6.RP.A.3a | Constructed response

6. 6 square meters; aligned pairs (0,0), (4,3), (8,6)

8 ÷ 4 = 2 batches; 2 × 3 = 6.

6.RP.A.3a | Constructed response

7. 8 square meters; aligned pairs (0,0), (6,2), (24,8)

24 ÷ 6 = 4 batches; 4 × 2 = 8.

6.RP.A.3a | Constructed response

8. Aligned pairs (0,0), (6,4), (12,8), (42,28); 28 m²

Each aligned jump adds 6 liters and 4 m². Both quantities scale by 7; the number lines measure different units.

6.RP.A.3a | Double number line and explanation

9. Aligned pairs (0,0), (7,6), (14,12), (42,36); 36 m²

Each aligned jump adds 7 liters and 6 m². Both quantities scale by 6; the number lines measure different units.

6.RP.A.3a | Double number line and explanation

10. 28 square meters; aligned pairs (0,0), (6,7), (24,28)

24 ÷ 6 = 4 batches; 4 × 7 = 28.

6.RP.A.3a | Constructed response